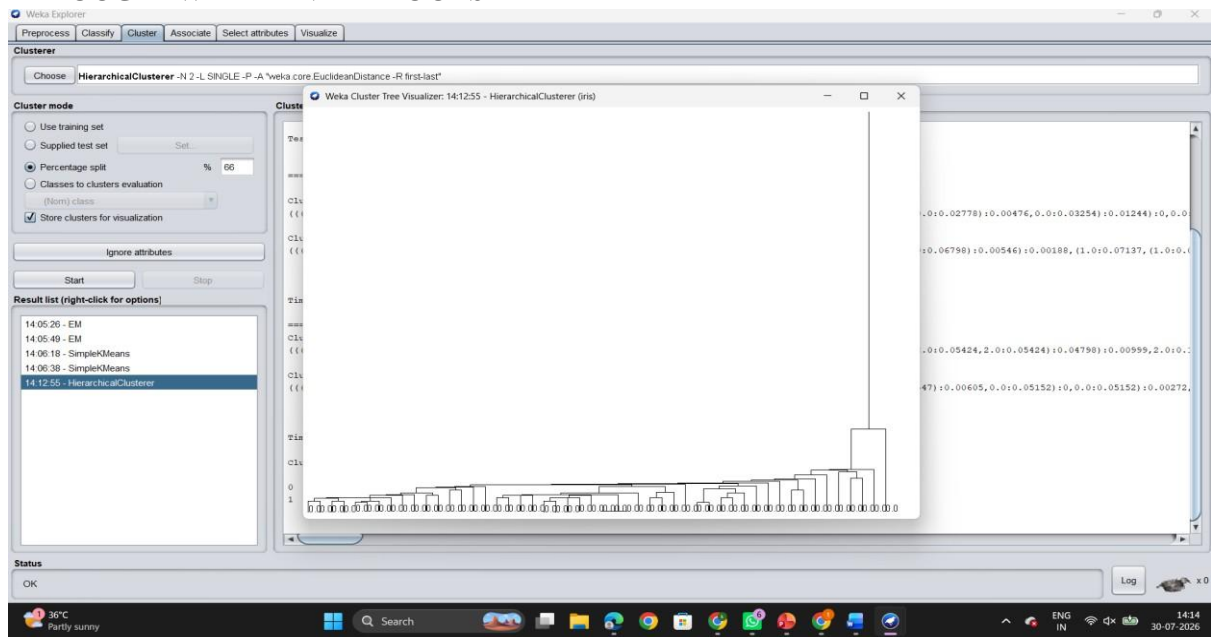


EXPERIMENT- 25 FREQUENT PATTERN MINING USING ASSOCIATION RULE THROUGH WEKA AND R -TOOLS



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start Stop

Result list (right-click):

- 14:21:52 - Apriori
- 14:22:54 - Apriori

Associator output

```
==== Associator model (full training set) ====
Apriori
=====
Minimum support: 0.15 (2 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 17

Generated sets of large itemsets:

Size of set of large itemsets L(1): 12
Size of set of large itemsets L(2): 47
Size of set of large itemsets L(3): 35
Size of set of large itemsets L(4): 6

Best rules found:

1. outlook=overcast 4 ==> play=yes 4 <conf:(1)> lift:(1.56) lev:(0.1) [1] conv:(1.43)
2. temperature=cool 4 ==> humidity=normal 4 <conf:(1)> lift:(0.14) [2] conv:(2)
3. humidity=normal windy=FALSE 4 ==> play=yes 4 <conf:(1)> lift:(1.56) lev:(0.1) [1] conv:(1.43)
4. outlook=sunny play=no 3 ==> humidity=high 3 <conf:(1)> lift:(2) lev:(0.11) [1] conv:(1.5)
5. outlook=sunny humidity=high 3 ==> play=no 3 <conf:(1)> lift:(2.6) lev:(0.14) [1] conv:(1.93)
6. outlook=rainy play=yes 3 ==> windy=FALSE 3 <conf:(1)> lift:(1.75) lev:(0.09) [1] conv:(1.29)
7. outlook=rainy windy=FALSE 3 ==> play=yes 3 <conf:(1)> lift:(1.56) lev:(0.08) [1] conv:(1.07)
8. temperature=cool play=yes 3 ==> humidity=normal 3 <conf:(1)> lift:(2) lev:(0.11) [1] conv:(1.5)
9. outlook=sunny temperature=hot 2 ==> humidity=high 2 <conf:(1)> lift:(2) lev:(0.07) [1] conv:(1)
10. temperature=hot play=no 2 ==> outlook=sunny 2 <conf:(1)> lift:(2.8) lev:(0.09) [1] conv:(1.29)
```

Status

OK Log

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start Stop

Result list (right-click):

- 14:21:52 - Apriori
- 14:22:54 - Apriori

Associator output

```
====
Minimum support: 0.15 (694 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 17

Generated sets of large itemsets:

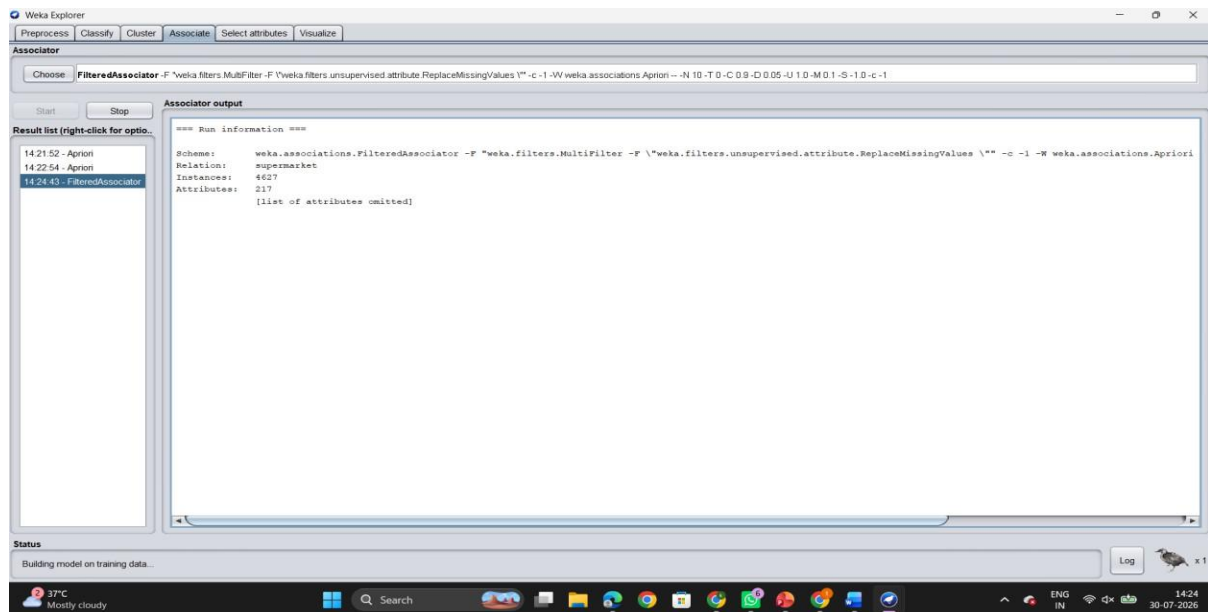
Size of set of large itemsets L(1): 44
Size of set of large itemsets L(2): 380
Size of set of large itemsets L(3): 910
Size of set of large itemsets L(4): 633
Size of set of large itemsets L(5): 105
Size of set of large itemsets L(6): 1

Best rules found:

1. biscuit= frozen foods= fruit= total=high 768 ==> bread and cake= 723 <conf:(0.92)> lift:(1.27) lev:(0.03) [155] conv:(3.35)
2. baking needs= biscuit= fruit= total=high 760 ==> bread and cake= 696 <conf:(0.92)> lift:(1.27) lev:(0.03) [149] conv:(3.28)
3. baking needs= frozen foods= fruit= total=high 770 ==> bread and cake= 705 <conf:(0.92)> lift:(1.27) lev:(0.03) [150] conv:(3.27)
4. biscuit= fruit= vegetables= total=high 815 ==> bread and cake= 746 <conf:(0.92)> lift:(1.27) lev:(0.03) [159] conv:(3.26)
5. party snack food= fruit= total=high 854 ==> bread and cake= 779 <conf:(0.91)> lift:(1.27) lev:(0.04) [164] conv:(3.15)
6. biscuit= frozen foods= vegetables= total=high 797 ==> bread and cake= 725 <conf:(0.91)> lift:(1.26) lev:(0.03) [151] conv:(3.06)
7. baking needs= biscuit= vegetables= total=high 772 ==> bread and cake= 701 <conf:(0.91)> lift:(1.26) lev:(0.03) [145] conv:(3.01)
8. biscuit= fruit= total=high 854 ==> bread and cake= 866 <conf:(0.91)> lift:(1.26) lev:(0.04) [179] conv:(3)
9. frozen foods= fruit= vegetables= total=high 834 ==> bread and cake= 757 <conf:(0.91)> lift:(1.26) lev:(0.03) [156] conv:(3)
10. frozen foods= fruit= total=high 969 ==> bread and cake= 877 <conf:(0.91)> lift:(1.26) lev:(0.04) [179] conv:(2.92)
```

Status

OK Log



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose Logistic - R 1.0E-8 - M -1 - num-decimal-places 4

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation Folds 10

☐ Percentage split % 66

More options...

(Nom) class

Start Stop

Result list (right-click for options)

- 14.30.59 - rules.ZeroR
- 14.32.05 - rules.ZeroR
- 14.32.58 - rules.ZeroR
- 14.36.18 - bayes.BayesNet
- 14.36.40 - trees.DecisionStump
- 14.36.52 - trees.J48
- 14.37.04 - lazy.KStar
- 14.37.13 - trees.RandomForest
- 14.37.33 - trees.RandomTree
- 14.41.21 - functions.SMO
- 14.44.30 - functions.Logistic
- 14.47.08 - functions.Logistic
- 14.47.28 - functions.Logistic

Classifier output

```
petallength 0 0.0001
petalwidth 0 0

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances 144 96 %
Incorrectly Classified Instances 6 4 %
Kappa statistic 0.94
Mean absolute error 0.0287
Root mean squared error 0.1424
Relative absolute error 6.456 %
Root relative squared error 30.2139 %
Total Number of Instances 150

=== Detailed Accuracy By Class ===

TP Rate FP Rate Precision Recall F-Measure MDC ROC Area PRG Area Class
1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 Iris-setosa
0.920 0.020 0.958 0.920 0.939 0.910 0.972 0.934 Iris-versicolour
0.960 0.040 0.923 0.960 0.941 0.911 0.972 0.934 Iris-virginica
Weighted Avg. 0.960 0.020 0.960 0.960 0.960 0.940 0.981 0.956

=== Confusion Matrix ===

a b c <-- classified as
50 0 0 | a = Iris-setosa
0 46 4 | b = Iris-versicolour
0 2 48 | c = Iris-virginica
```

Status

OK Log x 0

37°C Mostly cloudy

Search

ENG IN

14:47 30-07-2026